

The FFGFT Hawking Mechanism

KMS temperature, winding transition, $\mathbb{Z}_3$ triality selection, unitarity — the complete mechanism from FFGFT-native building blocks

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Main result (residual entropy at the endpoint): “ $S_{\text{BH}}(M_{\text{coll}})$ equals exactly Matzke’s stability threshold n_{thr} ” $\rightarrow S_{\text{BH}}(M_{\text{coll}}) = 2\pi/\ln 2 \text{ bit} = L_{\text{Matzke}}/\ell_{\text{P}}$, $n_{\text{thr}} = S_{\text{BH}}/\sqrt{2}$ (cf. Doc. 190, R144).
- Structural constant: $I_{\text{sector}}/I_{\text{thermal}} = \log_2 3 \approx 1.585$ (bits over nats) $\rightarrow \ln 3 \approx 1.099$ (same unit).

Abstract

Doc. 313 (Ch. G) established the thermodynamic frame of Hawking radiation in FFGFT: a single KMS rule tempers Unruh, Gibbons–Hawking and black holes jointly **[K]**, the membrane picture receives its thermometer **[S]**, and the family ladder reaches the cosmos at $r_s = R_H/2$ **[S]**. What remained open was the *microscopic emission mechanism*: which modes escape, what the quantum carries, why massive quanta fall back. This document works it out completely from FFGFT-native building blocks:

1. **Temperature:** The KMS rule $T = \hbar/(k_B\tau)$ with membrane period $\tau = 8\pi GM/c^3$ follows from $\tilde{T} \cdot m = 1$ (clock compression at the membrane, Doc. 313 G.2). **[K]**
2. **Quantum:** The Hawking quantum is the bit of scale $4\pi r_s$: $k_B T_H = \hbar c/(4\pi r_s)$ exactly — the system-dependent bit energy $E_{\text{bit}} = \hbar c/L$ (Doc. 257) evaluated at the horizon scale. No universal bit value is needed anywhere. **[K]**
3. **Selection:** Only triality-0 modes ($T_R = 0$) can be emitted. Confinement (Doc. 321) and Hawking selection are *one* principle; absorption of the partner quantum is the $\mathbb{Z}_3$ projector orthogonality $P_j P_k = 0$. **[B]**

4. **Carrier:** The massless carrier is the $n_4 = 0$ torus mode (spatial winding without mass-circle winding). Massive modes ($n_4 \neq 0$) have finite $\tilde{T}$ and fall back. **[K]**
5. **Information:** The emitted quantum carries the sector pair $(k, -k)$ as an orthogonally readable quantum number ($\log_2 3$ bits) — plus the thermal 1-nat area bookkeeping. Fine-grained entropy remains constant (unitarity, Doc. 322). No information paradox. **[K]**

Two verification scripts, all assertions passed: `325_hawking_ffgft_mechanismus.py` (thermodynamic chain), `325_hawking_z3_selektion.py` ($\mathbb{Z}_3$ algebra).

.1 Starting point: what Doc. 313 left open

Doc. 313 Ch. G established three results: (a) the single KMS rule $T = \hbar/(k_B \tau)$ for Unruh, Gibbons–Hawking and black holes **[K]**; (b) the membrane thermometer from $\tilde{T} \cdot m = 1$ **[S]**; (c) the family ladder up to the cosmos ($r_s = R_H/2$ exactly) **[S]**. Likewise the limit: evaporation is 56 (stellar) resp. 83 (supermassive) orders of magnitude too slow for the time cycle — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it **[K]**.

What was missing was the *microstructure* of emission:

- Which modes of the $T^4/\mathbb{Z}_3$ spectrum can leave the horizon — and which principle selects them?
- What is the emitted quantum, and how is its energy related to the information unit of the framework (Doc. 257)?
- Why do massive quanta fall back — as a consequence of the geometry, not as an extra assumption?
- How is information conservation realized microscopically?

These four questions are answered here — each from an already established FFGFT building block: $\tilde{T} \cdot m = 1$ (Docs. 077, 180), system-dependent bit energy (Docs. 257, 302, A271–A273), $\mathbb{Z}_3$ triality (Doc. 321), unitarity of the spectral operator (Doc. 322).

.2 Building block 1: the KMS temperature from $\tilde{T} \cdot m = 1$

Wherever the local time structure is periodic with period τ , an observer reads off temperature:

$$T = \frac{\hbar}{k_B \tau} \quad (1)$$

System	Period τ	$T = \hbar/(k_B \tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18} \text{ s}$	$2.66 \times 10^{-30} \text{ K}$
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4} \text{ s}$	$6.17 \times 10^{-8} \text{ K}$
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^8 \text{ s}$	$3.98 \times 10^{-20} \text{ K}$

The membrane period $\tau = 8\pi GM/c^3$ is not an extra assumption but follows from time-mass duality: mass compresses the time structure ($\tilde{T} \cdot m = 1$ — the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with exactly this period (Doc. 313 G.2). **[K]**

The temperature is thus not the result of a Bogoliubov computation on a curved background, but a read-off rule of the clock map — the same rule for all three systems.

.3 Building block 2: the Hawking quantum as a system-dependent bit

The central new result of this document. The system-dependent bit energy of FFGFT (Doc. 257) reads $E_{\text{bit}}(L) = \hbar c/L$. Evaluated at the horizon scale:

Core result [K]

$$k_B T_H = \frac{\hbar c}{4\pi r_s} = E_{\text{bit}}(L = 4\pi r_s) \quad (2)$$

The thermal Hawking quantum *is* the bit of scale $L = 4\pi r_s$ (twice the horizon circumference). Numerically exact (script 1): $\hbar c/(k_B T_H) = 4\pi r_s$ to machine precision.

The system dependence of bit values (Docs. 257, 302, A271–A273) is therefore no obstacle to Hawking physics but *its explanation*: the Hawking temperature is the bit temperature of the horizon scale. A universal bit value — say, a fixed bit mass from a Landauer evaluation at the Planck temperature — is needed nowhere; every system carries its own bit energy, and that of the black hole at its horizon is exactly $k_B T_H$.

Area bookkeeping

Emission of a quantum of energy $k_B T_H$ carries, by Clausius, exactly $dS = E/(T \cdot k_B) = 1$ nat, hence

$$dA = -4\ell_P^2 \text{ per nat} \quad (3)$$

(script 1). The area loss per emission thus follows from Bekenstein + Clausius alone — no further geometric assumptions. **[B]**

.4 Building block 3: $\mathbb{Z}_3$ triality selection

The algebraic core of the mechanism. The horizon carries the $\mathbb{Z}_3$ orbifold structure of $T^4/\mathbb{Z}_3$ (Docs. 321, 322). The three $\mathbb{Z}_3$ projectors

$$P_k = \frac{1}{3} \sum_{j=0}^2 \omega^{-jk} \tau^j, \quad \omega = e^{2\pi i/3} \quad (4)$$

are complete, idempotent and orthogonal (script 2):

$$P_0 + P_1 + P_2 = \mathbb{1}, \quad P_j P_k = \delta_{jk} P_j. \quad (5)$$

Absorption [B]

The infalling partner quantum erases the horizon voxel exactly when it lies in the orthogonal $\mathbb{Z}_3$ sector:

$$P_j (P_k \psi) = 0 \quad (j \neq k) \quad (6)$$

— exact, no tunnelling (script 2: $\|P_2(P_1 \psi)\| < 10^{-15}$). Pair creation at the membrane separates a $(k, -k)$ sector pair; the inward-falling partner is absorbed by projector orthogonality, the outward-running one escapes provided it satisfies the triality condition.

Who escapes: $T_R = 0$

The triality of a mode is its $\mathbb{Z}_3$ charge k . Emissible (asymptotically free) are only combinations with $T_R = \sum k_i \equiv 0 \pmod{3}$:

Combination	T_R	emissible
Single mode $k = 0$ (photon analogue)	0	yes
Single mode $k = 1$ (quark)	1	no
Bilinear (1, 2) (meson)	0	yes
Bilinear (1, 1)	2	no
Trilinear (1, 1, 1) (baryon)	0	yes

Confinement = Hawking selection [B]

Doc. 321 identified confinement as triality selection $T_R = 0$. The same principle acts at the horizon: only colourless objects escape. Confinement and Hawking selection are *one* principle of the $\mathbb{Z}_3$ structure — not two. The $\mathbb{Z}_3$ charge is exactly conserved in every emission (script 2).

.5 Building block 4: the massless carrier

On $T^4 = T_{\text{space}}^3 \times S_m^1$ the mass of a mode is carried by the winding n_4 on the mass circle ($\tilde{T} \cdot m = 1$). The emissible carrier is the mode with

$$n_4 = 0 : \quad \text{spatial winding without mass winding.} \quad (7)$$

A massive quantum ($n_4 \neq 0$) has finite $\tilde{T}_i$ in the compressed clock map at the membrane (Doc. 313 G.2) it falls back — as a consequence of the geometry, not as an extra assumption. For $|n_i| \leq 1$ there are $3^3 - 1 = 26$ massless neighbouring modes against 54 massive ones (script 2). **[K]**

The Hawking spectrum thus consists of $n_4 = 0$ modes with $T_R = 0$: massless, colourless, thermal at T_H — the photon-like degrees of freedom of the torus.

.6 Building block 5: information conservation

What the quantum carries

The emitted $T_R = 0$ quantum carries the *sector pair* $(k, -k)$ as an internal quantum number. The three possible pair states are exactly orthogonal (Gram matrix = $\mathbb{1}_3$, script 2) — which sector was withdrawn is readable from the quantum: $\log_2 3 \approx 1.585$ bits of sector information per quantum, in addition to the thermal 1-nat area bookkeeping.

Fine-grained constancy

Doc. 313 G.5 phrased it as “two languages, one statement”: locally, the radiation is correlated with the fractal membrane structure ($S_{\text{BH}}(M_\odot) = 1.05 \times 10^{77} k_B$ of correlations — nothing is lost); globally, fine-grained entropy is constant under unitary evolution (unitarity of the spectral operator, Doc. 322 **[K]**). No information paradox: the membrane is a clock-map object, not a causal abyss.

Quantitative information budget [B]

Building blocks 1–5 supply all ingredients for a fully computable information budget over the entire lifetime of a black hole.

Endpoint of evaporation. From Doc. 329 [B]: the collapse threshold $M_{\text{coll}} = m_{\text{p}}/\sqrt{2}$ follows from pure geometry (Compton = Schwarzschild). The Schwarzschild radius at the endpoint is

$$r_{\text{s}}(M_{\text{coll}}) = \sqrt{2} \ell_{\text{p}}, \quad (8)$$

exactly one Planck length times $\sqrt{2}$ — the smallest horizon on which a T^4 mode still fits.

Residual entropy at the endpoint — exact identity [B]:

Main result R93 [B]

$$S_{\text{BH}}(M_{\text{coll}}) = \frac{4\pi r_{\text{s}}^2}{4 \ell_{\text{p}}^2} \Big|_{r_{\text{s}}=\sqrt{2}\ell_{\text{p}}} = 2\pi \text{ nat} = \frac{2\pi}{\ln 2} \text{ bit} \approx 9.065 \text{ bit}. \quad (9)$$

This is $S_{\text{BH}}(M_{\text{coll}}) = 2\pi / \ln 2 \text{ bit} = L_{\text{Matzke}}/\ell_{\text{p}}$; Matzke's stability threshold is $n_{\text{thr}} = S_{\text{BH}}/\sqrt{2}$, where $\sqrt{2} = r_{\text{s}}(M_{\text{coll}})/\ell_{\text{p}}$ (Doc. 329 [B]) — the same geometry (Compton = Schwarzschild) in two languages. *(Corrected on 1 Oct. 2026: previously "This equals exactly Matzke's stability threshold n_{thr} " — $n_{\text{thr}} = 2\pi/(\sqrt{2} \ln 2) = 6.410$, whereas $2\pi / \ln 2 = 9.065 = L_{\text{Matzke}}/\ell_{\text{p}}$ (Doc. 329); the two differ by $\sqrt{2} = r_{\text{s}}(M_{\text{coll}})/\ell_{\text{p}}$, and the identity remains exact; cf. Doc. 190, R144 "R93 refined".)*

Total emission volume. For a black hole of initial mass $M_{\text{init}} > M_{\text{coll}}$:

$$N_{\text{emit}} = S_{\text{BH}}(M_{\text{init}}) - 2\pi \text{ nat}. \quad (10)$$

Example $M_{\odot}$: $N_{\text{emit}} \approx 1.05 \times 10^{77} \text{ nat}$.

Structural constant of the $\mathbb{Z}_3$ geometry [B]. The ratio of sector information to thermal information is identical for every mass:

$$\frac{I_{\text{sector}}}{I_{\text{thermal}}} = \frac{\log_2 3 \text{ bit}}{1 \text{ nat}} = \frac{\log_2 3}{1/\ln 2} = \ln 3 \approx 1.099, \quad (11)$$

since each emitted quantum carries exactly $\log_2 3$ bits of sector information (§.6) and $1 \text{ nat} = 1/\ln 2$ bits of thermal area bookkeeping. *(Corrected on 1 Oct. 2026: previously $\log_2 3/1 = \log_2 3 \approx 1.585$ — numerator in bits, denominator in nats; with $1 \text{ nat} = 1.4427$ bits the ratio is $1.585/1.4427 = \ln 3 = 1.0986$, still independent of mass and scale.)* The ratio is a structural constant of the $T^4/\mathbb{Z}_3$ geometry, independent of mass and scale. [R94 [B]]

Verification script: 325_informationsbilanz.py — 9 assertions, all passed.

.7 The FFGFT correction and its limits

Doc. 313 gives the fractal correction of the Hawking power:

$$P = P_{\text{std}} (1 - \xi \ln(M/m_{\text{p}})), \quad (12)$$

numerically 0.60 % (PBH 10^{12} kg) to 1.44 % (SMBH $10^9 M_{\odot}$) — script 1. It does not change the evaporation findings: only primordial holes below

$$M_{*} = \left(\frac{\tau_{\text{c}} \hbar c^4}{5120 \pi G^2} \right)^{1/3} = 3.3 \times 10^{11} \text{ kg} \quad (13)$$

close via Hawking within one time cycle $\tau_c = 91.9$ Gyr. The Sun is 56, an SMBH 83 orders of magnitude too slow (script 1) — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it (Doc. 313 G.4). **[K]**

The family ladder runs seamlessly up to the cosmos: $T_H = T_{GH}$ requires $M = c^3/(4GH_0) = 4.66 \times 10^{52}$ kg with $r_s = R_H/2$ exactly (script 1). **[K]**

Placement of the bit result

$k_B T_H = E_{\text{bit}}(4\pi r_s)$ shows: Hawking physics can be formulated entirely with the system-dependent bit energy. Any construction that instead postulates a universal bit value (say, a fixed bit mass from $k_B T \ln 2$ at a privileged temperature) adds a parametrization that the temperature physics does not require — consistent with A271–A273: Landauer's bound is a statement about carrier operations, not about abstract information.

.8 Status overview

Element	Status
KMS rule tempers Unruh/GH/BH jointly	[K] (Doc. 313)
Membrane period from $\tilde{T} \cdot m = 1$	[K]
$k_B T_H = E_{\text{bit}}(4\pi r_s)$ — quantum = horizon bit	[K] (new)
Area quantum $-4\ell_P^2/\text{nat}$ from Bekenstein+Clausius	[B] (new)
$\mathbb{Z}_3$ absorption $P_j P_k = 0$	[B] (new)
$T_R = 0$ selection; confinement = Hawking selection	[B] (new)
Carrier = n_4 -free torus mode	[K] (new)
Sector info $\log_2 3$ bits orthogonally readable	[B] (new)
Information conservation (membrane + unitarity)	[K] (313/322)
FFGFT power correction 0.6–1.4 %	[K] (313)
Greybody factors, mode-spectrum detail	[S] (open)
Back-reaction on orbifold fixed points	[S] (open)

The Hawking mechanism in FFGFT thus stands at **[K]/[B]**; open remain only the greybody factors (standard QFT computation on the FFGFT background) and the back-reaction of the emission on the orbifold fixed-point structure.

Verification scripts

- python/Dok325_Skripte/325_hawking_ffgft_mechanismus.py — thermodynamic chain: KMS, membrane thermometer, horizon bit, area bookkeeping, correction, evaporation, family ladder, information side.
- python/Dok325_Skripte/325_hawking_z3_selektion.py — $\mathbb{Z}_3$ algebra: projectors, absorption, triality selection, charge conservation, sector information, masslessness.

References

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